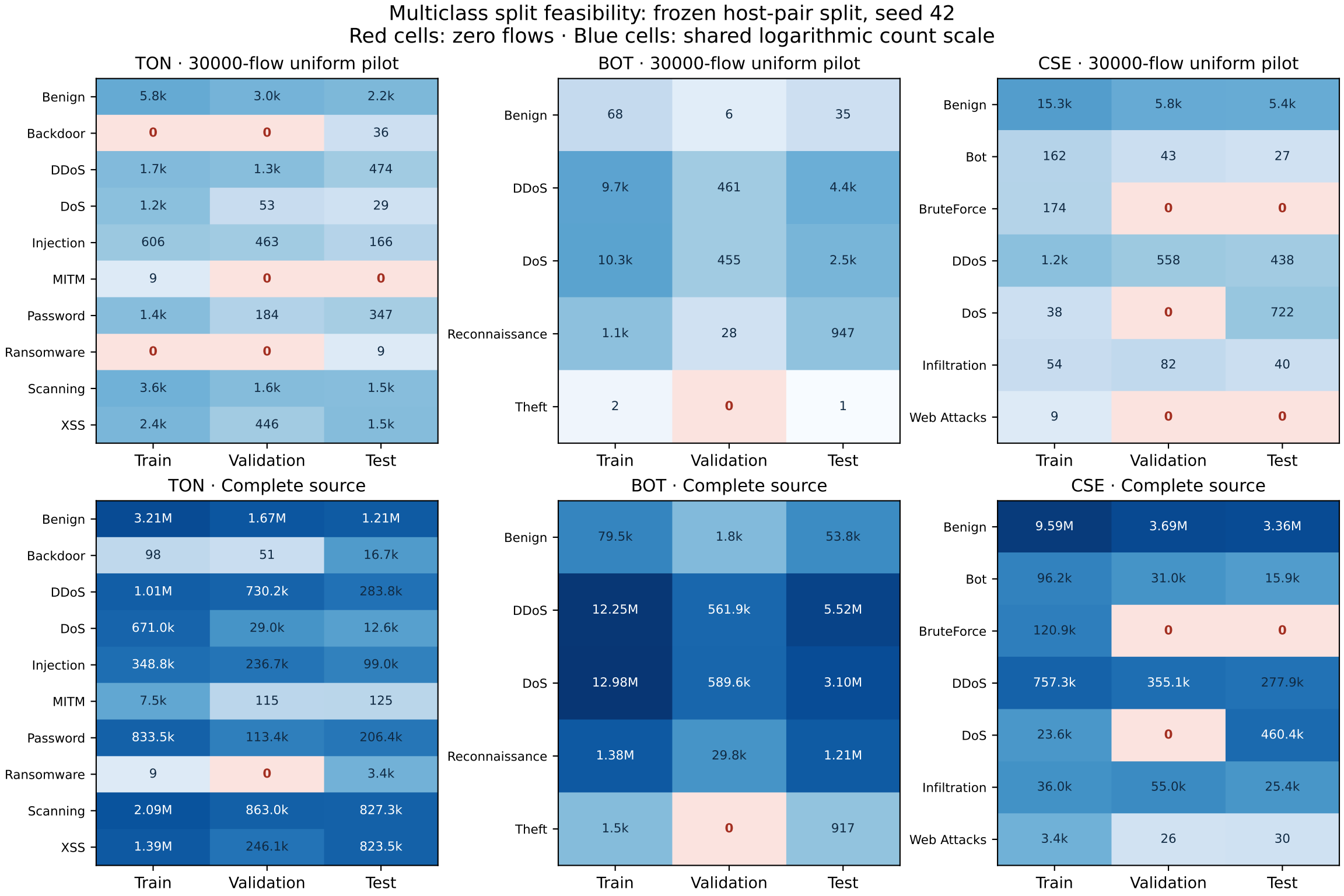
Multiclass split feasibility: frozen host-pair split, seed 42
Red cells: zero flows · Blue cells: shared logarithmic count scale
TON · 30000-flow uniform pilot
Benign 5.8k 3.0k 2.2k
Backdoor 0 0 36
DDoS 1.7k 1.3k 474
DoS 1.2k 53 29
Injection 606 463 166
MITM 9 0 0
Password 1.4k 184 347
Ransomware 0 0 9
Scanning 3.6k 1.6k 1.5k
XSS 2.4k 446 1.5k
Train Validation Test
BOT · 30000-flow uniform pilot
Benign 68 6 35
DDoS 9.7k 461 4.4k
DoS 10.3k 455 2.5k
Reconnaissance 1.1k 28 947
Theft 2 0 1
Train Validation Test
CSE · 30000-flow uniform pilot
Benign 15.3k 5.8k 5.4k
Bot 162 43 27
BruteForce 174 0 0
DDoS 1.2k 558 438
DoS 38 0 722
Infiltration 54 82 40
Web Attacks 9 0 0
Train Validation Test
TON · Complete source
Benign 3.21M 1.67M 1.21M
Backdoor 98 51 16.7k
DDoS 1.01M 730.2k 283.8k
DoS 671.0k 29.0k 12.6k
Injection 348.8k 236.7k 99.0k
MITM 7.5k 115 125
Password 833.5k 113.4k 206.4k
Ransomware 9 0 3.4k
Scanning 2.09M 863.0k 827.3k
XSS 1.39M 246.1k 823.5k
Train Validation Test
BOT · Complete source
Benign 79.5k 1.8k 53.8k
DDoS 12.25M 561.9k 5.52M
DoS 12.98M 589.6k 3.10M
Reconnaissance 1.38M 29.8k 1.21M
Theft 1.5k 0 917
Train Validation Test
CSE · Complete source
Benign 9.59M 3.69M 3.36M
Bot 96.2k 31.0k 15.9k
BruteForce 120.9k 0 0
DDoS 757.3k 355.1k 277.9k
DoS 23.6k 0 460.4k
Infiltration 36.0k 55.0k 25.4k
Web Attacks 3.4k 26 30
Train Validation Test